

Mixtum: a graphical tool for two-way admixture analysis in population genetics based on f -statistics

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Abstract

Summary: *Mixtum* is a Python-based code that estimates ancestry contributions in a process of two-way admixture based on bi-allelic genotype data. The outcomes of *Mixtum* come from the geometric interpretation of the f -statistics formalism. Designed with user-friendliness as a priority, *Mixtum* allows to interactively handle a menu of user-supplied populations to build different mixture models in conjunction with the set of auxiliary populations required by the framework. The results are presented graphically and numerically. Importantly, *Mixtum* provides a novel index (an angle) that assesses the quality of the ancestral reconstruction of the model under scrutiny. The use and interpretation of the outcomes of *Mixtum* are explained and illustrated with case studies.

Availability and implementation: The open source code is available on GitHub at <https://github.com/jmcastelo/mixtum> and on Zenodo at <https://doi.org/10.5281/zenodo.17789375>. *Mixtum* is implemented in Python and runs on Linux, Windows and macOS.

1 Introduction

Given a triplet of populations characterized by Single Nucleotide Polymorphisms (SNP), *Mixtum* determines whether two of them can be the parental populations, or ancestry sources, of the third, which represents the result of an admixture process. We aim to provide this tool as a user-friendly method to evaluate admixture scenarios where the main mathematical difficulty stems from the fact that the sampled proxies are genetically distant from the ancestral triplet, as they have accumulated the effect of genetic drift. The so-called f -statistics formalism (Patterson *et al.* 2012, Haak *et al.* 2015, Lazaridis *et al.* 2016, Peter 2016, Lipson 2020, Harney *et al.* 2021, Oteo-García and Oteo 2021, Peter 2022, Oteo and Oteo-García 2024) is a method that reduces the evolutionary genetic drift contributions and provides estimates of the mixing coefficient.

Mixtum implements part of a geometric interpretation of the methodological framework originally introduced in Oteo-García and Oteo (2021) and Oteo and Oteo-García (2024). What follows is a brief overview of its mathematical foundation. For

additional details and mathematical reasoning behind the method, readers should refer to the original publications.

Each population is interpreted as a point in a space in which the axes represent the allele frequencies of the SNPs genotypes. The dimension of this frequency space is the number of SNPs in the dataset. At the beginning of an admixture process the three points are aligned, with the admixed population in between, closer to the main donor in case of asymmetric contributions. The relative distances are in correspondence with the admixture proportion α . After that, genetic drift causes these three points to follow Brownian-like trajectories over time and to acquire a triangular disposition, which is the situation expected with the proxy data. The key of f -statistics stays in the approximate restoration of the ancestral alignment, removing the effect of genetic drift in the three populations. To this end, a number of auxiliary populations are required in the dataset, which are also referred to as outgroups or reference populations in the literature.

Any misalignment in the allele frequency space can be quantified by the angle at the admixed population vertex. This is done in two scenarios. We can measure: (i) an angle ϕ (termed pre-JL)

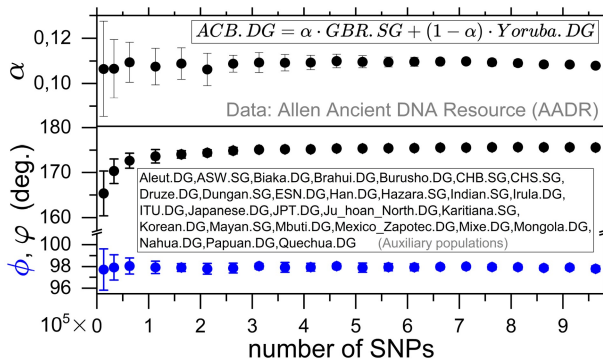


Figure 2 Dependence on the number of SNPs of the post-JL α and ϕ (top), and the pre-JL ϕ (bottom).

space. Each point is the average of fixed-size blocks of adjacent SNPs, whose origins are randomly located in the SNP sequence. The error bar is one standard deviation.

3 How to Mixtum

In the *Input files* tab the triad of data files is loaded and then parsed and checked for a valid structure. A populations file can be loaded and modified later on in the *Populations* tab. The number of SNPs to be processed can be limited. The allele frequencies table is then built up and can be saved in plain text format with the populations as columns and their labels in the header. The number of processors may be adjusted to compute in parallel. In the *Admixture model* tab, the model is set. The remaining populations may be selected at will to act as auxiliary, with a lower bound equal to four; or eight, if bootstrap is enabled. All the numerical outcomes can be exported. The task is then to interpret the results:

- Regression plot tab.** (i) The slope (1) (color bar) has to be $\alpha \in [0, 1]$, for a model to be valid. Consider the two error bars: the first from the dispersion of points in the linear fit and the second from the bootstrap on auxiliary populations. (ii) Outliers in the plot: removing certain auxiliary populations (click on the points to identify them) refines the determination of α and ϕ .
- Angles tab.** (i) The closer ϕ is to 180° , the better the reconstruction of the ancestral alignment. The error bar comes from bootstrapping auxiliary populations. (ii) $\phi < \phi$ indicates reduction of the effect of genetic drift. (iii) The conventional admixture test is $\phi > 90^\circ$ (which implies $f_3(\alpha, b; x) < 0$), but there may be cases $\phi < 90^\circ$ which, after JL projection, appear as admixture scenarios, i.e. $90^\circ < \phi$. (iv) A model is ruled out whenever $\phi > \phi$.
- f_4 -ratio histogram tab.** Obtained from (1) with $m = 2$, the f_4 -ratio is a well-known statistic related to the estimation of α . This distribution is usually quite dispersed. Some auxiliary populations may give values out-of-range. This tab is intended to illustrate the potential misinterpretation from using single values of the f_4 -ratio, not as a tool to measure α .
- PCA tab.** The set of populations to be projected can be selected from those in the allele frequency table. An admixture triplet tends to appear aligned in a PCA projection (Peter

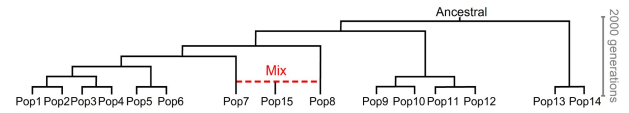


Figure 3 Simulated phylogenetic graph (*msprime*).

2022). It can be also used to gain insight about the auxiliary populations that generate outliers in the fit.

- f -statistics tab.** Specific values of f_2 , f_3 and f_4 , can be computed.
- Output logs.** Main results log includes: (i) admixture model, number of SNPs and auxiliary populations, (ii) α given with 95% CI error bars obtained from the linear fit (the first) and from the bootstrap on auxiliary populations (the second). (iii) ϕ without error estimate and ϕ with 95% CI error bar from auxiliary populations bootstrap, and (iv) below, tidbits: the pre-JL admixture proportion, a statistical average of (in range) f_4 -ratios and the f_3 admixture test.

4 Case studies

Three case studies are presented below. The first is based on the simulated phylogeny in Fig. 3 (data available with the code) generated with *msprime* (Baumdicker et al. 2022). The next two are cases with real data from the Allen Ancient DNA Resource (AADR) V9 (Mallick et al. 2024), with 1 113 154 SNPs. The outcomes are gathered in Table 1 as well as in the panels of Fig. 4.

1. *msprime* simulated phylogeny

The dataset has 15 populations with 10 individuals each and 85 761 SNPs. We study the model $Pop15 = \alpha \cdot Pop7 + (1 - \alpha) \cdot Pop8$, which has nominal value $\alpha = 0.2$. In addition, we have studied the variants: $Pop15 = \alpha \cdot Pop7 + (1 - \alpha) \cdot Any$, and $Pop15 = \alpha \cdot Any + (1 - \alpha) \cdot Pop8$. In all cases the 12 free populations act as auxiliaries. The outcomes have been ranked according to the value of ϕ and the first ones presented in Table 1 (upper block), where the nominal case leads. Note that none of the pre-JL values of ϕ , smaller than 90° , initially point toward a mixing scenario unless we consider the possibility of a large effect of genetic drift. The linear fits are in Fig. 4 (top). Δ_1 and Δ_2 stand for 95% CI obtained from the linear fit (the first) and from the bootstrap on auxiliary populations (the second).

Using 850 300 SNPs the nominal admixture proportion is accurately reproduced, with ϕ close to 180° [last line (*) of the upper block of Table 1]. The effect of increasing the number of SNPs is also observed in the ranges of the regression plot in Fig. 4 (top), for the nominal case.

2. $ACB.DG = \alpha \cdot GBR.SG + (1 - \alpha) \cdot Yoruba.DG$

This is a well-known historical example of an admixed group in the form of the Afro-Caribbean population (ACB), as an example to validate it as the product of admixture between a European source (represented by the *GBR* population as a proxy) and an African source (represented by the *Yoruba* population as a proxy). Twenty seven auxiliary populations have been chosen (see inset in Fig. 2).

The result for the slope α in Fig. 4 is the commonly accepted one. The estimates for both mixing angles in Table 1, $\phi \gg \phi > 90^\circ$, confirm that the model corresponds to a real

Table 1 Two-way admixture models.^a

Mix	Donor 1	Donor 2	Angle (degrees)		$\alpha \pm \Delta_1 \pm \Delta_2$
			ϕ	$\varphi \pm \Delta_2$	
Pop15	Pop7	Pop8	80	149±18	0.18±0.03±0.05
	Pop6	Pop8	77	132±35	0.20±0.04±0.10
	Pop1	Pop8	76	118±42	0.27±0.09±0.15
	Pop7	Pop10	66	69±97	0.16±0.20±0.30
	Pop7	Pop14	68	66±84	0.20±0.24±0.36
	Pop14	Pop8	71	39±30	-0.30±0.19±0.69
	Pop7*	Pop8*	81	170±5	0.20±0.01±0.02
ACB	GBR.SG	Yoruba.DG	97	175±3	0.108±0.001±0.007
	Han.DG	Brahui.DG	49	27±18	-0.5±0.2±0.8

^a Top: simulated data. Nominal models are boldfaced. Number of SNPs is 85 761, except 850 300 in the last line (*). Bottom: Real data from with 1 113 154 SNPs.

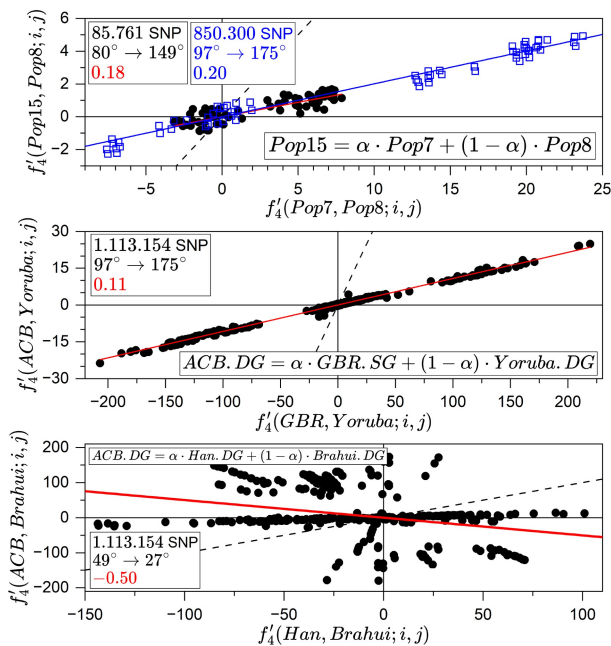


Figure 4 Linear fits for case studies. The legends contain information about the admixture model, the number of SNPs, the pre-JL and post-JL angles, and the slope of the fit (solid lines). Auxiliary populations i, j , are those in Fig. 2. The dashed line has a slope equal to one.

mixture, with a large reduction in genetic drift. Furthermore, φ close to 180° indicates that the sources are close to the exact ones.

If instead we choose the combination: $\text{Yoruba.DG} = \alpha' \cdot \text{GBR.SG} + (1 - \alpha') \cdot \text{ACB.DG}$, then the resulting linear fit (not shown) is as good as the model above, with slope: $\alpha' = -\alpha/(1 - \alpha) = -0.12$; and $\phi = 72^\circ > \varphi = 4^\circ$. Thus, a good regression plot with slope out of range can hide a potentially valid triplet in an alternative combination.

3. $\text{ACB.DG} = \alpha \cdot \text{Han.SG} + (1 - \alpha) \cdot \text{Brahui.DG}$

This model is unrealistic because neither *Han* nor *Brahui* are close to the actual parentals, which is illustrated by the distribution of points in Fig. 4 where the linear regression is not even

necessary. In any case, the confirmation can be found in Table 1: negative slope and $\varphi < \phi$.

5 Conclusion

Mixtum offers assistance in the study of two-way admixture in population genetics within the framework of f -statistics. It allows interactive testing of different models using arbitrary sets of auxiliary populations, tracking the role of each one. The estimate of the admixture proportions is buttressed by the value of the angle that attests to the reconstruction quality of the ancestral alignment of the model, a novel indicator in the f -statistics formalism. Along with the visual environment, this is the main difference from the available f -statistics packages.

Author contributions

José-María Castelo (Software [lead], Visualization [lead], Writing—review & editing [supporting]), José-Angel Oteo (Conceptualization [lead], Formal analysis [lead], Investigation [lead], Methodology [lead], Software [supporting], Visualization [supporting], Writing—original draft [lead], Writing—review & editing [lead]), and Gonzalo Oteo-García (Conceptualization [lead], Formal analysis [lead], Investigation [lead], Methodology [lead], Software [supporting], Supervision [lead], Visualization [supporting], Writing—original draft [supporting], Writing—review & editing [supporting])

Conflict of interests

None declared.

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Data availability

The curated dataset of human genotypes from Allen Ancient DNA Resource (AADR), v62.0_1240k, is available at <https://doi.org/10.7910/DVN/FFIDCW>. The simulated dataset is available with the *Mixtum* software at <https://doi.org/10.5281/zenodo.17789375> and <https://github.com/jmcastelo/mixtum>.

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